Semester II (First year] Branch/Course
Mechanical Engineering

Sl no	Category Index	Paper Code	Name of the Paper	Periods per week				Credit
				L	T	P	Total	
1	ESC	CSEUGES01	Programming for Problem Solving	3	0	0	3	3
2	ESC	EENUGES01	Basic Electrical Engineering	3	0	0	3	3
3	BSC	MATUGBS02	Engineering Mathematics II	4	0	0	4	4
4	BSC	CHMUGBS01	Engineering Chemistry	3	0	0	3	3
5	HSMC	ENGUGHU01	Communicative English	3	0	0	3	3
6	ESC	CSEUGES02	Programming for Problem Solving Lab	0	0	4	4	2
	ESC	EENUGES01	Basic Electrical Engineering Lab	0	0	3	3	1.5
8	ESC	MENUGES02	Workshop practice	0	1	3	3	2
9	BSC	CHMUGBS02	Engineering Chemistry lab	0	0	3	3	1.5
10	HSMC	ENGUGHU0	Language lab	0	0	2	2	1
Total Credits								**24**
Total Period per week								**31**

Semester II (First year)

ESC	**CSEUGES01**	**Programming for Problem Solving**	**3L-0T-0P**	**3 Credits**

Course outcomes:

Module	Content	Lecture
Module 1	**Introduction to computing:** block architecture of a computer, bit, bytes, memory, and representation of numbers in memory.	**1**
Module 2	**Introduction to problem solving:** Basic concepts of an algorithm, program design methods, flowcharts.	**1**
Module 3	**Introduction to C programming:** A Brief History of C, C is middle-level Language, is a Structured Language, Complier Vs Interpreters, The Form of a C Program, Library & Linking, Compilation & Execution process of C Program . [2]	**2**
Module 4	**Variables, Data Types, Operator & Expression:** Character Set, Token, Identifier & Keyword, Constant, Integer, Floating Point, Character, String, Enumeration, Data Types in C, Data Declaration & Definition Operator & Expression, Arithmetic, Relational, Logical, Increment & Decrement, Bit wise, Assignment, Conditional, Precedence & Associability of Operators.	**3**
Module 5	**Console I/O:** ntroduction, Character input & Output, String Input & Output, Formatted Input/Output (scanf/printf), sprintf & sscanf.	**2**
Module 6	**Control Statement:** Introduction, Selection Statements, Nested if, if-else-if, The “?” Alternative, The Conditional Expression, switch, Nested switch, Iteration Statements, for loop, while loop, do-while loop, Jump Statements, Goto & label, break & continue, exit() function.	**4**
Module 7	**Array & String:** Single Dimension Arrays, Accessing array elements, Initializing an array, Multidimensional Arrays, Initializing the arrays, Memory Representation, Accessing array elements, String Manipulation Functions, searching, sorting an array.	**6**
Module 8	**Function:** Introduction, advantages of modular design, prototype declaration, Arguments & local variables, Returning Function Results by reference & Call by value, passing arrays to a function, Recursion.	**4**

Module 9	**Storage Class & Scope:** Meaning of Terms, Scope - Block scope & file scope, Storage Classes Automatic Storage, Extern Storage, Static, Storage, Register Storage.	**2**
Module 10	**Pointers:** Introduction, Memory Organization, The basics of Pointer, The Pointer operator Application of Pointer, Pointer Expression, Declaration of Pointer, Initializing Pointer, De-referencing Pointer,Void Pointer, Pointer Arithmetic, Precedence of &, * operators Pointer to Pointer, Constant Pointer, Dynamic memory allocation, passing pointer to a function, array of pointers, accessing arrays using pointers, handling strings using pointers.	**4**
Module 11	**Structure, Union, Enumeration & typedef:** Structures, Declaration and Initializing Structure, Accessing Structure members, Structure, Assignments, Arrays of Structure, Passing, Structure to function, Structure Pointer, Unions.	**2**
Module 12	**C Preprocessor:** Introduction, Preprocessor Directive, Macro Substitution, File Inclusion directive, Conditional Compilation.	**2**
Module 13	**File handling:** Introduction, File Pointer, Defining & Opening a File, Closing a File, Input/Output Operations on Files, Operations on Text mode files and binary mode files, Error Handling During I/O Operation, Random Access To Files, Command Line Arguments	**3**

Suggested Books:

1. B.S. Gottfried: Programming in C; TMH.
2. B.W. Kernighan and D.M. Ritchie: The C Programming Language; PHI.
3. H. Schildt: C++: The Complete Reference; TMH, 4e.
4. B. Stroustrup: The C++ Programming Language; Addison-Wesley.
5. E. Balagurusamy: Programming in ANSI C; TMH.
6. Yashwant Kanetkar: Let Us C; BPB Publications.
7. K. N. King: C Programming: A Modern Approach, W. W. Norton and Company. Pradip Dey and Manas Ghosh: Programming in C, Oxford University Press

ESC	**EENUGES01**	**Basic Electrical Engineering**	**3L-0T-OP**	**3Credits**

Course outcomes:

Module	Content	Lecture
Module 1	**Introduction:** Basic concepts of Electrostatics and Electromagnetic.	**4**
Module 2	**DC Circuit**: Introduction of Electric Circuit & Elements, Loop Analysis, Node analysis, Star (Y) - Delta (Δ) & Delta (Δ)-Star (Y) Transformations.	**6**
Module 3	**DC Network Theorem**: Superposition Theorem, Thevenin's theorem, Norton's theorems, Maximum Power Transfer Theorem, Reciprocity Theorem, Time-domain analysis of first-order RL and RC circuits.	**8**
Module 4	**Single-phase AC Circuits** Generation of Sinusoidal Voltage Waveform (AC) and Some Fundamental Concepts, Representation of Sinusoidal Signal by a Phasor, Analysis of single-phase ac circuits consisting of R, L, C, RL, RC, RLC combinations (series and parallel), resonance. Three phase balanced circuits, voltage and current relations in star and delta connections.	**8**
Module 5	**Transformer:** Definition, working principle & construction, EMF equation, Equivalent circuit, Open circuit & Short circuit tests, Efficiency & Regulation.	**4**
Module 6	**DC Machines**: Constructional Features of D.C Machines , Principle of Operation of D.C Machines, EMF & Torque Equation , D.C Generators, D.C Motors, Losses, Efficiency, 3-point Starter and speed control of DC shunt Motor.	**4**
Module 7	**Three-phase Induction Motor**: Introduction to 3-phase induction motor	**1**
Module 8	**Introduction to Power System:** Basic concepts of Power System	**1**

Text book:

BSC	**MATUGBS02**	**Engineering Mathematics II**	**4L-0T-0P**	**4 Credits**

Course Objective:

Module	Content	Lecture
Module 1	Matrices: Matrix operations (Addition, Multiplication, Transpose), invertible matrix.	**4**
Module 2	Determinant and their properties.	**2**
Module 3	Row reduced echelon form; Rank of a matrix. Solution of the matrix equation Ax = b; Cramer's rule. Eigenvalues and eigenvectors, characteristic polynomial of a matrix, Cayley–Hamilton theorem and its application. Linear dependence and independence of vectors, basis and dimension. Complex numbers and Complex integrals. Inequalities, Theory of equations.	**32**
Module 4	Complex numbers and Complex integrals. Inequalities, Theory of equations.	**18**
Module 5	Differential equation of first order and first degree: Exact, separable and homogeneous differential equations, Bernoulli's equation, ODEs of first order but not of first degree; Clairaut's equation.	**7**
Module 6	Higher order linear equation with constant coefficients: Complementary function, Particular integral, Symbolic Operator D.	**4**
Module 7	Method of undetermined coefficients, Euler's homogeneous equation and deduction to an equation of constant coefficients.	**4**
Module 8	Second order linear equation with variable coefficients: exact equation: reduction of order; variation of parameters; reduction to normal form; change of independent variables. Simple eigenvalue problems.	**4**

Module 9	System of linear differential equations with constant coefficients.	**2**

References:

1. Advanced Engineering Mathematics : Erwin Kreyszig
2. Advanced Engineering Mathematics : R.K. Jain & S. R. K lyengar
3. Advanced Engineering Mathematics : C. R. Wylle & L. C. Barrett
4. Differential & Integral Calculus : N. Plskunov

BSC	**CHMUGBS01**	**Engineering Chemistry**	**3L-0T-0P**	**3Credits**

Course Outcomes:

Module	Content	Lecture
Module 1	**Thermodynamics :** Importance and scope, definitions of system and surroundings; type of systems; Extensive and intensive properties; Steady state and equilibrium; Zeroth law of thermodynamics; First law of thermodynamics, internal energy and Enthalpy as a state function; Second law of thermodynamics; Kelvin, Planck and Clausius statements; Carnot cycle and refrigerator; Carnot's theorem; Physical concept of entropy.	
Module 2	**Water and its treatment :** Sources of water, Impurities in water, Hardness of water, Determination of hardness of water, Water quality parameter, Treatment of water for domestic purpose, Waste water.	
Module 3	**Polymers :** Terminology, Classification of polymers, Polymerization techniques, Molecular weight of polymers, Plastics, Rubbers, Fibers, Conducting and semiconducting polymers, Natural polymers.	
Module 4	**Green Chemistry :** Definition and concept of green chemistry, Emergence of green chemistry, Alternative solvents, Design of safer chemicals, Microwave radiation of green synthesis, Green laboratory Technology.	

Books referred

1. K. S. Maheswaramma and M. Chugh, Engineering Chemsitry, Pearson, 2016.

2. Wiley Engineering Chemistry, Wiley, 2nd Edn., 2014.

HSMC	**ENGUGHU01**	**Communicative English**	**3L-0T-0P**	**3Credits**

Objectives of the Course: To impart basic Communication skills to the first year UG students in the English language through rigorous practice and use of various categories of common words and their application in sentences; to enable them to achieve effective language proficiency for their social, professional & inter personal communication both in speaking & writing.

Module	Content	Lecture
Module 1	**Fundamentals of Communication:** Communication: Meaning, Nature, Process, Importance and Function of Communication; Levels of Communication: Intra-personal, Interpersonal, Organizational, Mass Communications; The Flow of Communication: Downward, Upward, Lateral or Horizontal, Diagonal, Grapevine Communication; Network in an Organization; Principles for Effective Communication; Verbal and Non-Verbal Communication; Barriers to Communication, Gateways to Communication.	
Module 2	**Listening and Speaking Skills:** The Process of Listening; Barriers to Listening; Types of Listening: Active and Passive Listening; Methods for improving listening skills, Benefits of Effective Listening. Presentation Strategies: Defining Purpose; Organizing Contents; Preparing Outline; Audio-visual Aids; Nuances of Delivery; Body Language; Dimensions of Speech – Accent, Pitch, Rhythm, Intonation, Strong and Weak Forms, Connected Speech- Assimilation and Elision, Paralinguistic Features of Voice; Articulation of Speech Sounds- Vowels and Consonants; Spelling and Pronunciation; Problems of Indian speakers of English and their remedial measures.	
Module 3	**Reading and Writing Skills :** **Reading Skills:** Purpose, Process, Methodologies, and Strategies; Special Reading Situations – Skimming and Scanning, Intensive and Extensive Reading, Critical Reading, Drawing Inferences, Reading Technical Reports, etc. **Writing Skills:** Words and Phrases: Word Formation, Synonyms and Antonyms, Homophones, One Word Substitutes, Words Often Confused, Word Choice - Right Words, Appropriate Words, Idioms and Phrases; Correct Usage: Parts of Speech, Modals, Concord, Articles, Infinitives, Requisites of Sentence Construction. Elements of Effective Writing, Main Forms of Written	

	Communication: Paragraph - Techniques and Methods (Inductive, Deductive, Linear, Spatial, Chronological etc.), The Art of Condensation- various types (Précis, Summary and Abstract, etc.), Description, Agenda, Minutes, Notices, Circulars, Memo, Advertisements, Drafting an E-mail, Press Release.	
Module 4	Business Communication: Business Letters: Principles; Sales & Credit letters; Claim and Adjustment Letters; Job application and Résumés. Reports: Types; Significance; Structure, Style & Writing of Reports. Technical Proposal; Parts; Types; Writing of Proposal. Negotiation & Business Presentation skills.	

Suggested Readings:.

1. Sethi, J & et al. *A Practice Course in English Pronunciation*, Prentice Hall of India, New Delhi.
2. Berry Cicely: *Your Voice and How to Use it Successfully*, George Harp & Co. Ltd, London
3. Bansal, R.K. and J.B. Harrison. *Spoken English*, Orient Longman.
4. Hornby's, A.S. *Oxford Advanced Learners Dictionary of Current English*, 7th Edition.
5. Pillai, Sabina & Agna Fernandez: *Soft Skills & Employability Skills.* Cambridge Univ. Press.
6. Sudharshana, N.P. & C. Savitha: *English for Technical Communication*, Cambridge Univ. Press.
7. Raman, Meenakshi & Sangeeta Sharma: *Technical Communication: Principles and Practice.* Oxford Univ. Press.
8. Prasad, P. *The Functional Aspects of Communication Skills*, Delhi.
9. McCarthy, Michael. *English Vocabulary in Use*, Cambridge University Press, Cambridge.
10. Leech, G & Svartvik, J. *A Communicative Grammar of English*. Pearson Education. New Delhi.
11. Narayanaswamy V.R. *Strengthen your Writing*. Orient Longman, London.
12. Dean, Michael. *Write it*, Cambridge University Press, Cambridge.
13. Sen, Leena. *Communication Skills*, Prentice Hall of India, New Delhi. Bown, G. *Listening and Spoken*

English , Longman, London

ESC	**CSEUGES02**	**Programming for Problem Solving Lab**	**0L-0T-4P**	**2 Credits**

Course Outcomes:

Module	Content	Lecture
Module 1	Primary goal of this course is to make acquaint the students to know the programming language and also to know how 'C' can be used to write serious program to solve the problems. Programs will be based on the theoretical paper and to cover the concept of basic arithmetic operations, control statements, functions, recursions, arrays, strings, pointers, structures, unions, file handling etc.	

Suggested Books:

1. B.S. Gottfried: Programming in C; TMH.
2. B.W. Kernighan and D.M. Ritchie: The C Programming Language; PHI.
3. H. Schildt: C++: The Complete Reference; TMH, 4e.
4. B. Stroustrup: The C++ Programming Language; Addison-Wesley.
5. E. Balagurusamy: Programming in ANSI C; TMH.
6. Yashwant Kanetkar: Let Us C; BPB Publications.
7. K. N. King: C Programming: A Modern Approach, W. W. Norton and Company.
8. Pradip Dey and Manas Ghosh: Programming in C, Oxford University Press

ESC	**EENUGES01**	**Basic Electrical Engineering Lab (CSE,CEN & MEN)**	**3L-0T-0P**	**3 Credits**

Course Outcomes:

Module	Content	Practical
Module 1	Verification of Thevenin's Theorem	**3**
Module 2	Verification of Norton's Theorem	**3**
Module 3	Verification of Superposition Theorem	**3**
Module 4	Power Measurement of Fluorescent Lamp	**3**
Module 5	V-I characteristics of Incandescent Lamp	**3**
Module 6	Speed Control of DC motor Using Field and Armature Control	**3**

	Method	
Module 7	Starting and reversing of DC motor	**3**
Module 8	Open circuit and Short circuit test of Single Phase Transformer	**3**
Module 9	Calibration of Voltmeter and Ammeter	**3**
Module 10	Characteristics of Series R-L-C Circuit	**3**
Module 11	Characteristics of Parallel R-L-C Circuit	**3**
Module 12	Resistance measurement and continuity test of DC motor using Megger	**3**

ESC	**MENUGES02**	**Workshop Practice**	**0L-1T-2P**	**2 Credits**

Module	**Content**	**Lecture**
Module 1	Introduction to various hand tools e.g. allen keys, spanners, punch, files, hacksaw, hammers, chisels, vices, marking block, angle plates, etc.	
Module 2	Introduction to basic instruments: Vernier Caliper, Micrometer, Tri-square, Surface Plate, Height Gauge, Vernier Bevel Protractor, Screw Pitch Gauge, Radius Gauge, etc.	
Module 3	Demonstration on different machines and Equipments: Lathe, Milling, Drilling, Shaping, Radial Drilling, Grinding, Welding, Power Saw, Power Press, Planer Machine, Microscope, Profile Projector, etc.	
Module 4	Practical Exercises: Exercises involving the following operations: measuring and marking, sawing, chipping, filing, maintaining of perpendicularity of all surfaces by filing, making of taper surface by filing, making of curved surface by filing, plain turning, step turning and drilling.	

Reference books

1. Hazra Choudhury & Hazra Choudhury – Elements of Workshop Technology, Vol. I & II – Media Promoters and Publishers Pvt. Ltd.
2. Rajender Singh - Introduction to Basic Manufacturing Process and Workshop Technology, New Age International.

BSC	**CHMUGBS02**	**Engineering Chemistry Lab**	**0L-0T-3P**	**1.5 Credits**

Course Outcomes:

Module	Content	Lecture
Module 1	Acidimetric estimation of Sodium Carbonate and Sodium bi-Carbonate in their mixture.	**3**
Module 2	Estimation of Total Hardness of water by Complexometric method	**3**
Module 3	Estimation of Fe^{II} in Mohr's Salt by Permanganometric Titration.	**3**
Module 4	Qualitative analysis of single solid organic compounds.	**3**

HSMC	**ENGUGHU02**	**Language Lab**	**0L-0T-2P**	**1 Credits**

Course outcomes:

Objectives of the Course: To impart basic Communication skills to the first year UG students in the English language through rigorous practice and use of various categories of common words and their application in sentences; to enable them to achieve effective language proficiency for their social, professional & inter personal communication both in speaking & writing; to improve their English pronunciation

Module	Content	Lecture
Module 1	Group Discussion: Practical based on Accurate and Correct Grammatical Patterns.	
Module 2	Conversational Skills under suitable Professional Communication Lab conditions with emphasis on Kinesics: Interview, Greeting and Introducing, Leave taking, Asking Questions and Giving Replies, Inviting Friends and Colleagues, Negotiating, Persuading, Taking Initiatives, Praising and Complementing People, Expressing Sympathy, Seeking and Giving Permission, Complaining and Apologizing, Official/Public Speaking, Telephoning etc.	
Module 3	Communication Skills for Seminars/Conferences/Workshops with emphasis on Paralinguistic/ Kinesics.	
Module 4	Presentation Skills for Technical Paper/Research Paper/Project Reports/ Professional Reports based on proper Stress and Intonation Mechanics	
Module 5	Extempore, Argumentative Skills, Role Play Presentation with Stress and Intonation.	